11/11/2026 — MC / TD METHODS, CONCLUSIONS ON TABULAR RL

$q_{\pi}(s,a)$

PROBLEM OF EXPLORATION:

(1) EXPLORING STARTS

(2) ON-POLICY METHODS: THE POLICY WE FOLLOW TO PERFORM EXPERIMENTS IS PROGRESSIVELY IMPROVED.

→ IN POLICY IMPROVEMENT, WE CONSIDER THE <u>ε-GREEDY POLICY</u>

GPI : $V(S) \leftarrow V(S) + \alpha\left[G(S) - V(S)\right]$, $\forall S$ VISITED IN THE EPISODE

(3) OFF-POLICY LEARNING:

- π_b BEHAVIOR POLICY: WE USE π_b TO INTERACT WITH THE ENVIRONMENT. TO DO THE EXPERIMENTS TO COLLECT THE TRAINING DATA. MUST ENSURE EXPLORATION (TO HAVE ∞-MANY SAMPLES OF RETURN FROM ANY (s,a)).

- π_t <u>TARGET POLICY</u>: THE POLICY THAT PROGRESSIVELY CONVERGES TO π^*. CAN BE DETERMINISTIC.

$$q^* \longrightarrow \pi^*(s) = \underset{a}{\operatorname{argmax}}\, q^*(s,a) \quad \text{(DETERMINISTIC)}.$$

POLICY EVALUATION IN OFF-POLICY RL SETTING.

$\pi \xrightarrow{E} v_\pi$? USING DATA COMING FROM ANOTHER POLICY π_b

$$v_\pi(s) \triangleq E_\pi\{G_t \mid S_t = s\}$$

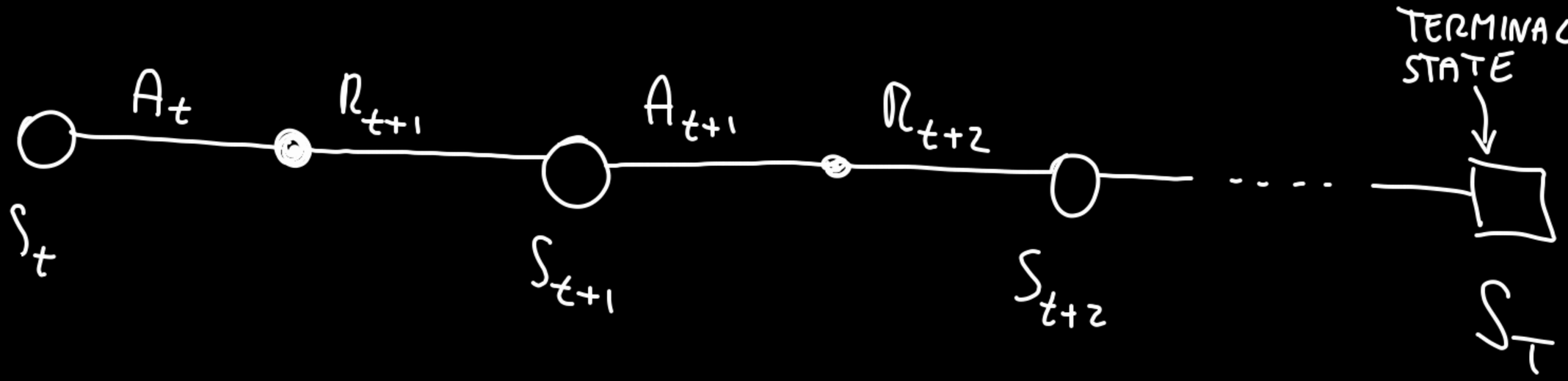
DOING THE AVERAGE OF THE RETURNS OBSERVED WHILE DOING EXPERIMENTS WITH π_b, THEY WILL CONVERGE TO

$v_{\pi_b}(s)$, NOT $v_\pi(s)$.

$$v_\pi(s) \triangleq E_\pi\{G_t \mid S_t = s\} = ?$$

HOW TO COMPUTE IN TERMS OF TRAJECTORIES τ AND ASSOCIATED $G(\tau)$?

CONSIDER THE GENERIC TRAJECTORY OBSERVED IN A GENERIC EPISODE.

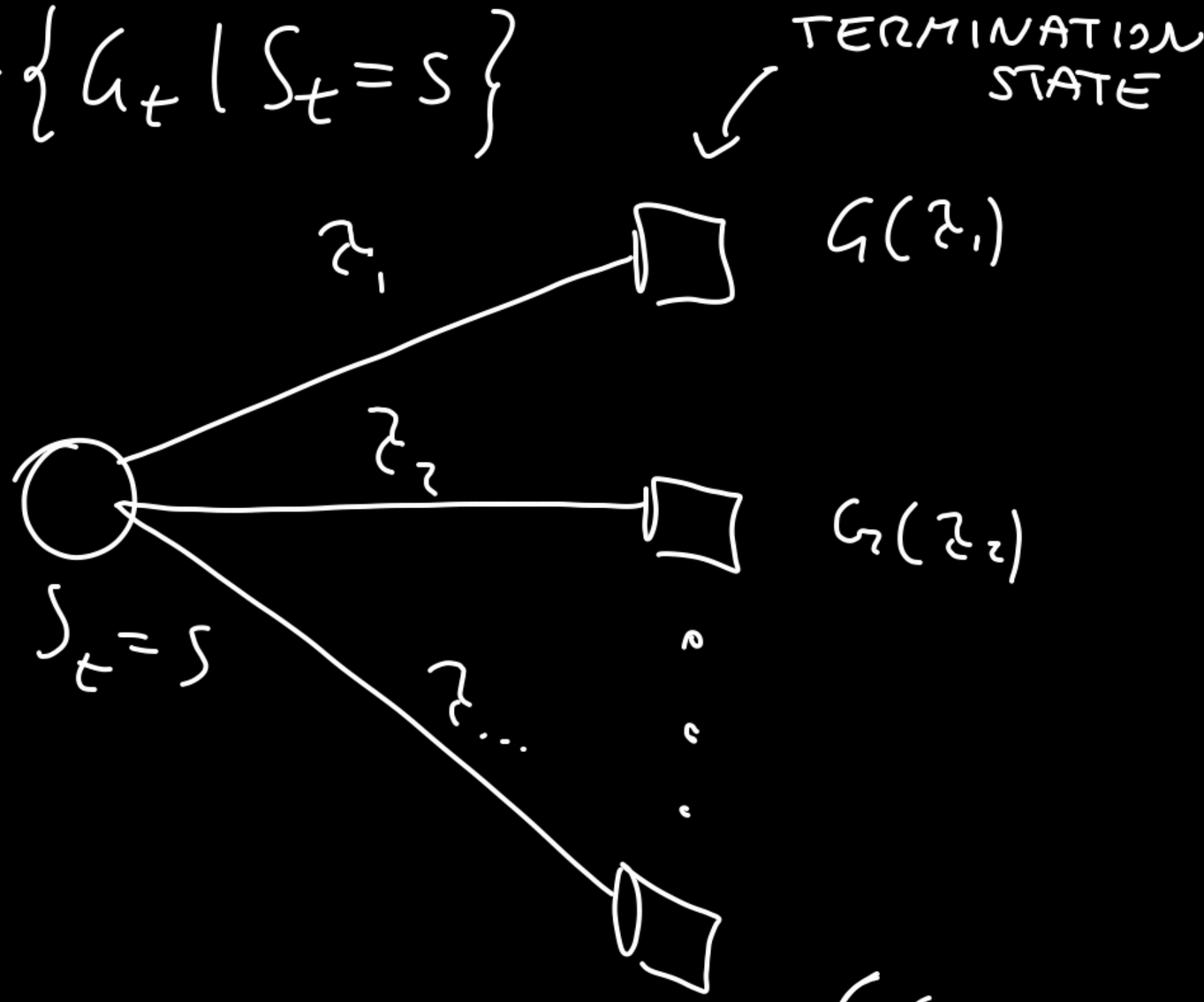


$$\tau = (S_t, A_t, R_{t+1}, S_{t+1}, \dots, S_T) \longrightarrow G(\tau)$$

SAMPLE TRAJECTORY

SAMPLE DISC. RETURN FOR SAMPLE TRAJ. τ.

$$\bar{E}_{\pi}\{G_t \mid S_t = s\}$$

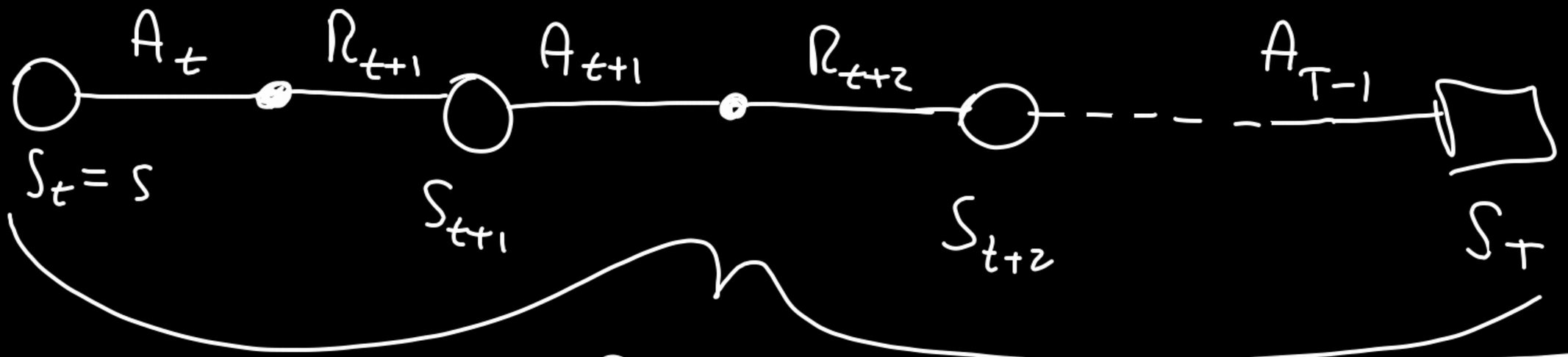


ALL POSSIBE DIFFERENT OUTCOMES OF EXPERIMENTS

$$E_\pi\{G_t \mid S_t = s\} = \sum_\tau P(\tau)\cdot G(\tau)$$

$$\| \quad v_\pi(s)$$

PROBABILITY OF OBSERVING τ.



$P(\tau) = ?$

$p(s', r \mid s, a)$

BUT p IS NOT KNOWN!

$$P(\tau) = \pi(A_t \mid S_t)\cdot p(S_{t+1} \mid S_t, A_t)\cdot \pi(A_{t+1} \mid S_{t+1}) \cdots p(S_T \mid S_{T-1}, A_{T-1})$$

$$P(\tau) = \prod_{k=t}^{T-1} \pi(A_k \mid S_k)\cdot p(S_{k+1} \mid S_k, A_k)$$

COMPACT FORMULA

$$V_\pi(s) = \sum_\tau P_\pi(\tau) \cdot G(\tau)$$

THIS IS WHAT WE WANT TO ESTIMATE.

($P_\pi(\tau)$ PROBABILITY OF OBSERVING τ UNDER POLICY π)

π_b - BEHAVIO POLICY USED TO GENERATE TRAJECTORIES.

→ DATA

$$V_{\pi_b}(s) = \sum_\tau P_{\pi_b}(\tau) \cdot G(\tau)$$

$$V_\pi(s) = \sum_\tau \underbrace{P_\pi(\tau) \cdot \frac{P_{\pi_b}(\tau)}{P_{\pi_b}(\tau)}}_{\triangleq \rho(\tau)} \cdot G_1(\tau) \quad , \forall s.$$

IMPORTANCE SAMPLING RATIO

π_b MUST SATISFY THE COVERAGE PROPERTY:

- $\pi(a|s) > 0 \longrightarrow \pi_b(a|s) > 0$

OTHERWISE WE DON'T HAVE DATA TO EVACUATE $V_\pi(s,a)$

$$V_\pi(s) = \underbrace{\sum_{\tau} P_{\pi_b}(\tau) \left(\rho(\tau) \cdot G(\tau) \right)}_{} \quad ; \forall s$$

THIS IS THE EXP. VALUE UNDER π_b, OF $\rho(\tau) G(\tau)$.

SO WE CAN EVALUATE $V_\pi(s)$ BY DOING AVERAGES OF THE WEIGHTED SAMPLE RETURN.

$$\rho(\tau) = \frac{P_\pi(\tau)}{P_{\pi_b}(\tau)} = \frac{\prod_{k=t}^{T-1} \pi(A_k|S_k) \cdot \cancel{P(S_{k+1}|S_k, A_k)}}{\prod_{k=t}^{T-1} \pi_b(A_k|S_k) \cdot \cancel{P(S_{k+1}|S_k, A_k)}}$$

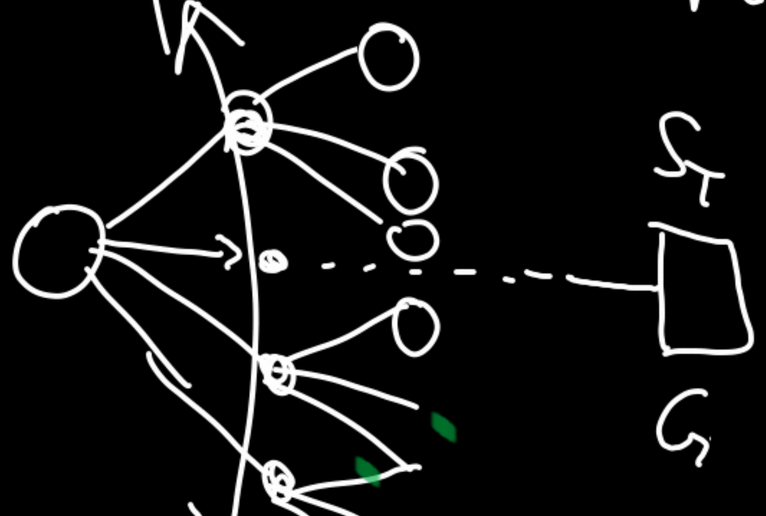
CAN BE COMPUTED. IT DEPENDS ONLY ON THE POLICIES.

ONE POSSIBILITY IS TO TAKE π_b AS ε-GREEDY POLICY.
AND π AS THE GREEDY POLICY.

CONCLUSION ON MC METHODS.

- SAMPLE UPDATES ~~OF~~ V/q (VERSUS THE EXPECTED UPDATES IN DYNAMIC PROGRAMMING METHODS).

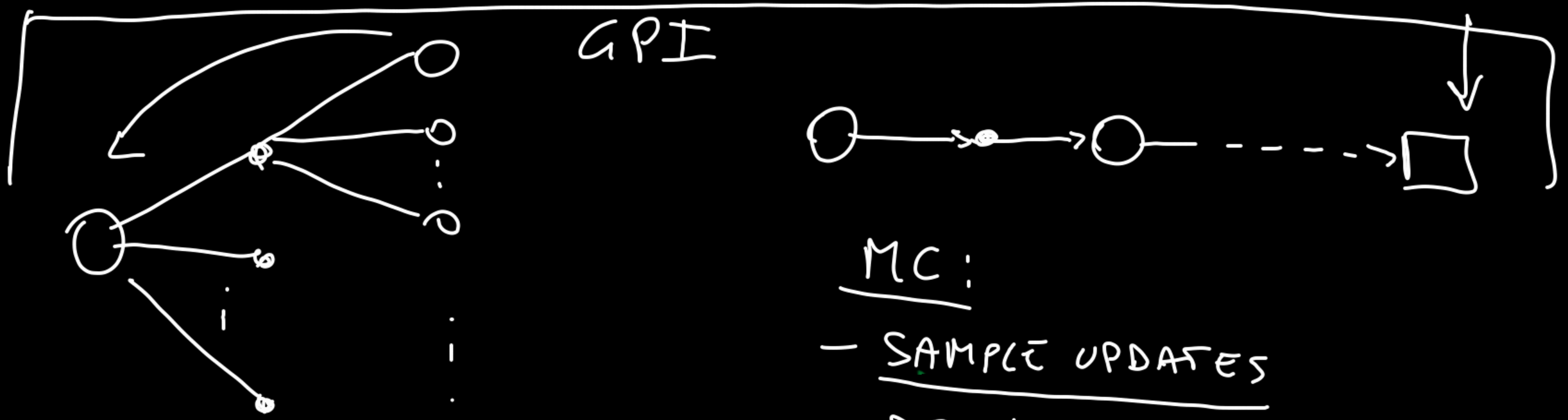
$$V(S) \leftarrow V(S) + \alpha [G - V(S)]$$



SAMPLE RETURN (THE ONE OBSERVED IN THE EPISODE).

(ITERATIVE WAY TO COMPUTE AN AVERAGE:
- SIMPLE AVERAGE
- EXP. AVERAGE
- MOVING AVERAGE)

IN MC METHODS, WE HAVE TO WAIT THE END OF EVERY EPISODE BEFORE WE CAN DO JUST 1 UPDATE



DYNAMIC PROGRAMMING:

- NEEDS p
- EXPECTED UPDATE
- BOOTSTRAP.

MC:

- SAMPLE UPDATES
- DON'T BOOTSTRAP
- LEARN FROM INTERACTION WITH ENVIRONMENT.

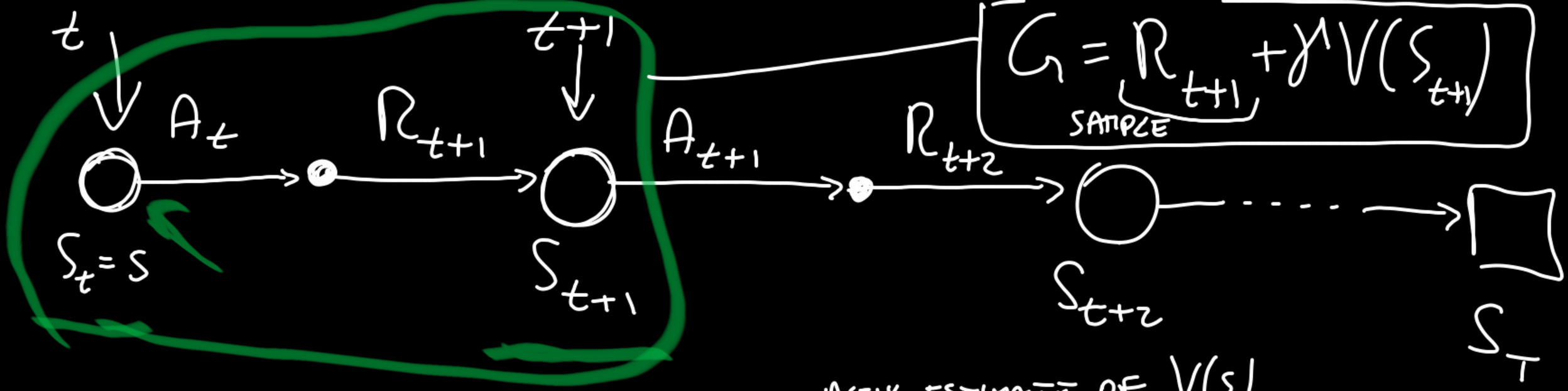
TEMPORAL DIFFERENCE (TD)

$\pi \rightarrow v_\pi$

POLICY EVALUATION: (DO WE NEED TO WAIT UNTIL END OF EPISODE TO BE ABLE TO UPDATE $V(S)$ – THE CURRENT ESTIMATE OF $v_\pi(s)$?)

CONSIDER A TRAJECTORY:

$(S_t = s, A_t, R_{t+1}, S_{t+1}, A_{t+1}, R_{t+2}, S_{t+2}, \ldots, S_T)$



MC UPDATE $V(s) \leftarrow$ NEW ESTIMATE OF $V(s)$ $\overbrace{V(s) + \alpha[\underbrace{G}_{\text{ESTIMATION ERROR}} - V(s)]}^{\text{OLD ESTIMATE}}$ (NO BOOTSTRAP)

$$v_\pi(s) = E_\pi\{G_t \mid S_t = s\}$$

$$v_\pi(s) = E_\pi\{R_{t+1} + \gamma G_{t+1} \mid \dots\}$$

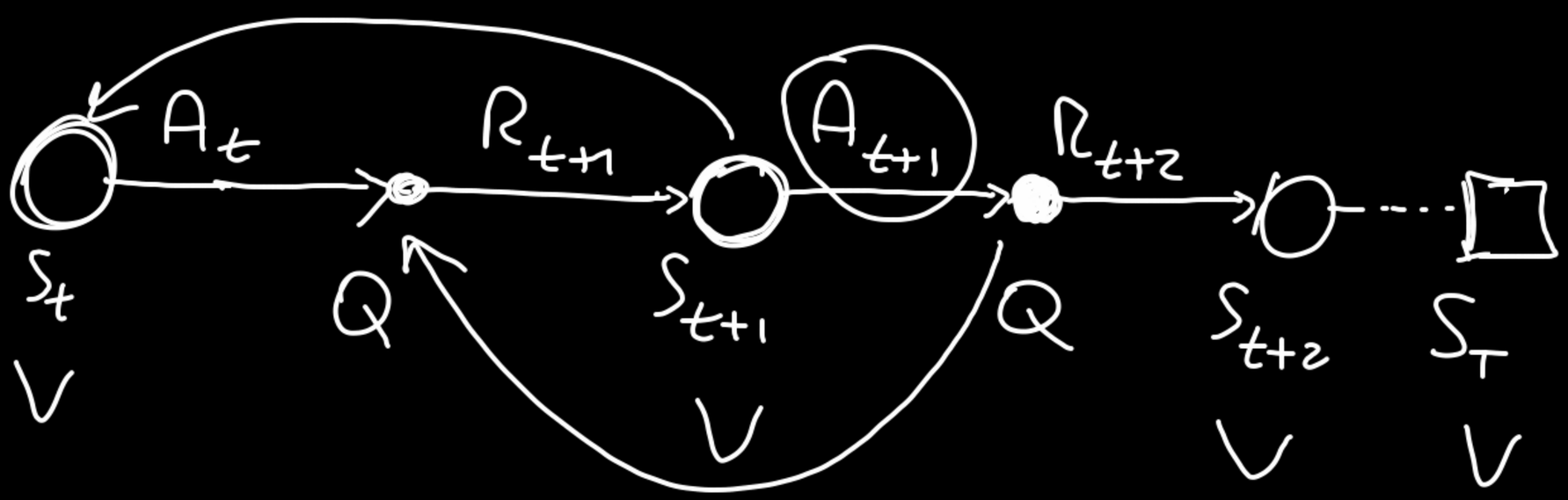
AT $t+1$, HOW CAN WE UPDATE $V(S_t)$?

TD(0) PREDICTION

$$V(S_t) \leftarrow V(S_t) + \alpha\left[\underbrace{R_{t+1} + \gamma V(S_{t+1})}_{\text{TARGET UPDATE VALUE}} - V(S_t)\right]$$

$V(S_T) = 0.$

HOW TO USE SAME CONCEPT FOR ESTIMATING q_π ?



$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\left[G_t - Q(S_t, A_t)\right]$$

SARSA ESTIMATION ALGORITHM.

$$\boxed{Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\left[\left(R_{t+1} + \gamma \underline{Q(S_{t+1}, A_{t+1})} - Q(S_t, A_t)\right]\right]}$$

$$G_t \cong R_{t+1} + \gamma V(S_{t+1}) \quad \text{OR} \quad G_t \cong R_{t+1} + \gamma Q(S_{t+1}, A_{t+1})$$

SARSA RL ALGORITHM (POLICY CONTROL: ITERATIVE EVALUATION AND IMPROVEMENT).

POSSIBLE GPI SARSA RL SCHEME:

$\forall$ EPISODE:

SELECT STARTING STATE

LOOP UNTIL S_T

- TAKE ACTION A_t ACCORDING TO ε-GREEDY POLICY, OBSERVE NEXT STATE S_{t+1} AND REWARD R_{t+1}.
- TAKE ANOTHER ACTION A_{t+1} ACCORDING TO ε-GREEDY POLICY.
- UPDATE $Q(S_t, A_t) \Leftarrow$ (UPDATE FORMULA OF PREVIOUS SLIDE).

ε-GREEDY (Q)

$$S_t \longrightarrow A^* = \underset{a}{\operatorname{argmax}}\, Q(S_t, a) \quad \begin{pmatrix} \text{GREEDY} \\ \text{ACTION} \end{pmatrix}$$

$$\pi^{\varepsilon\text{-GREEDY}}(S_t, a) = \begin{cases} \dfrac{\varepsilon}{|A(S_t)|} & \text{IF } a \neq A^* \\ 1 - \dfrac{\varepsilon}{|A(S_t)|} \cdot (|A(S_t)| - 1) & \text{IF } a = A^* \end{cases}$$

$(|A(S_t)| - 1)$: # OF NON GREEDY ACTIONS

$$\varepsilon \longrightarrow 0 \qquad \pi^{\varepsilon\text{-GREEDY}} \longrightarrow \pi^* \quad \begin{pmatrix} \text{GREEDY DET.} \\ \text{POLICY} \end{pmatrix}$$